EXPERIMENT : 12

AIM:

Experiment to detect outliers in a given data set.

DESCRIPTION:

Understand the procedure to identify the outliers in a given dataset

CODE:

```
import numpy as np

array = np.random.randint(1, 100, 16)

array
array.mean()
np.percentile(array, 25)
np.percentile(array, 50)
np.percentile(array, 75)
np.percentile(array, 100)

# Outlier detection
def outDetection(array):
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)
print(lr, ur)

import seaborn as sns
import matplotlib.pyplot as plt
```

```
# Distribution plot
sns.displot(array)
plt.show()

# Distplot (deprecated in newer Seaborn versions)
# Use this if your installed Seaborn version supports it
sns.distplot(array)
plt.show()

# Remove outliers
new_array = array[(array >= lr) & (array <= ur)]
print(new_array)

sns.displot(new_array)
plt.show()

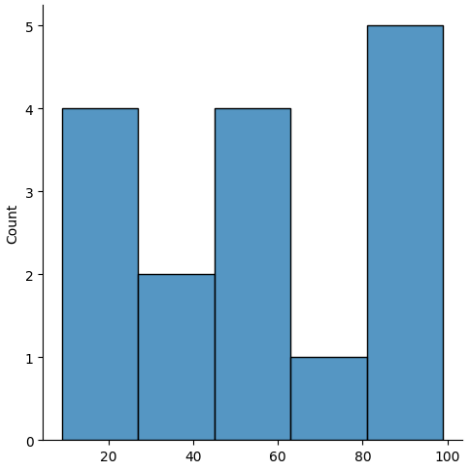
# Detect outliers again
lr1, ur1 = outDetection(new_array)
print(lr1, ur1)

final_array = new_array[
    (new_array >= lr1) & (new_array <= ur1)
]
print(final_array)

sns.distplot(final_array)
plt.show()
```

OUTPUT:

-51.75 166.25



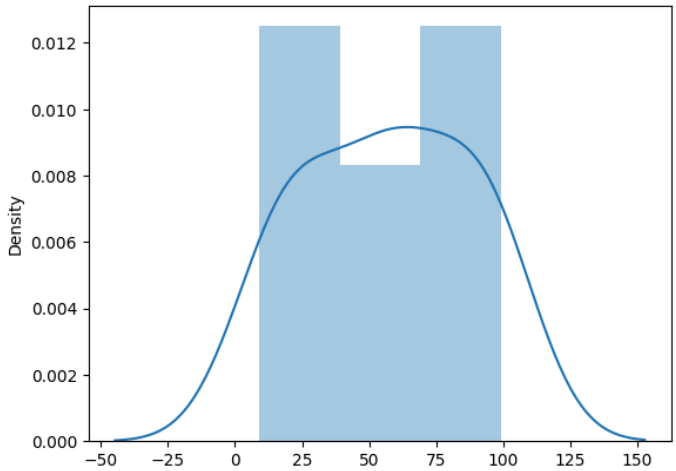
/tmp/ipykernel_2452/2256120119.py:32: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

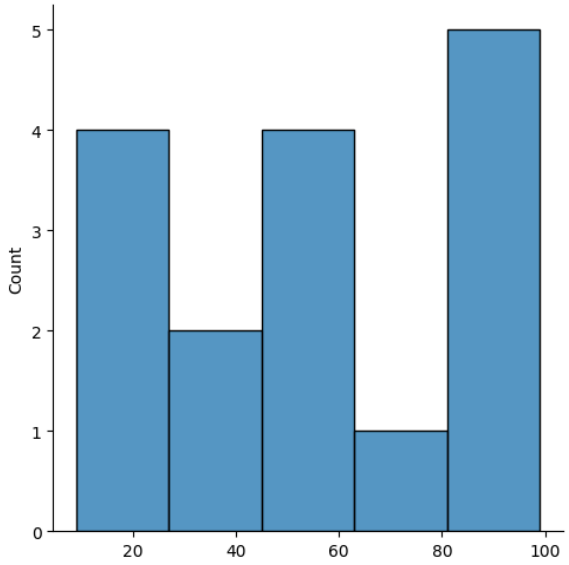
Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

```
sns.distplot(array)
```



[59 97 59 24 89 16 17 38 55 99 58 73 97 9 32 83]



-51.75 166.25
[59 97 59 24 89 16 17 38 55 99 58 73 97 9 32 83]

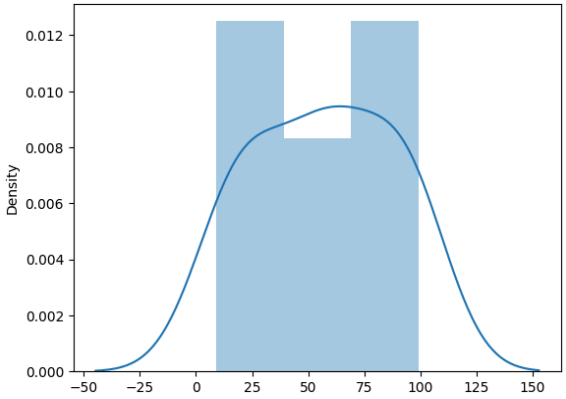
/tmp/ipykernel_2452/2256120119.py:51: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

```
sns.distplot(final_array)
```



RESULT:

Outliers in the given dataset were successfully identified using suitable statistical methods. The detection process helped identify unusual values and understand their impact on data analysis.